

WAVES · LIGHT

Why diamonds sparkle and straws look broken

The internet crosses oceans as flashes of light trapped inside glass threads thinner than a hair. A diamond throws light back at you from inside. A magnifying glass makes an ant look like a monster. All of it comes down to how light bounces and bends.

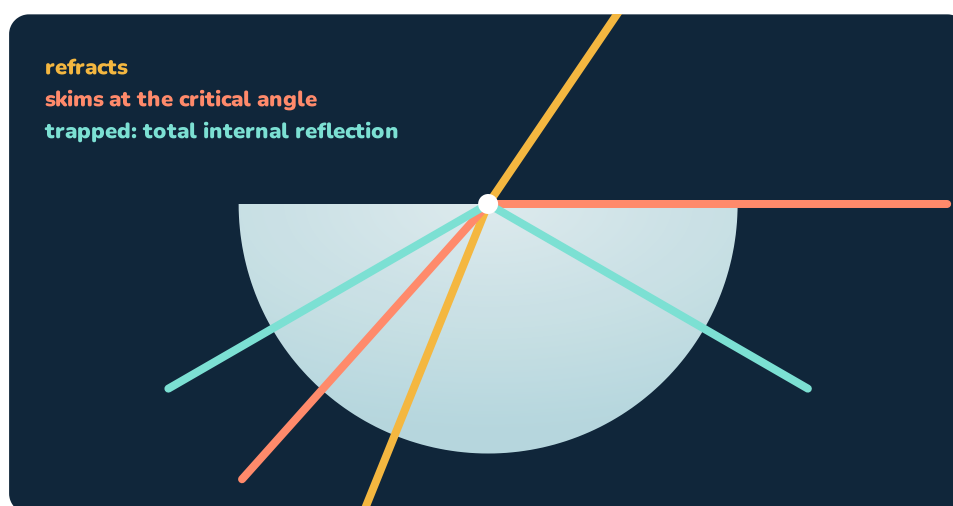
Sec 3–4 · Express

Exam weight: high

5 worked examples

15 practice questions

3 ray diagrams to draw



SEAB 6091 LEARNING OUTCOMES · BY THE END YOU CAN

1. use the terms **normal**, **angle of incidence** and **angle of reflection**, and apply $i = r$ in constructions, measurements and calculations
2. use the terms **normal**, **angle of incidence** and **angle of refraction** to describe refraction
3. recall and apply $\sin i / \sin r = \text{constant}$
4. define **refractive index** as the ratio of the speed of light in a vacuum to the speed in the medium
5. explain **critical angle** and **total internal reflection**, and apply them to **optical fibres** in telecommunication and medicine
6. describe the action of a **thin converging lens** on a beam of light, and define its **focal length**
7. draw **ray diagrams** for real and virtual images, and describe their characteristics

Name _____

Class _____

Date _____

01

Predict 1

Answer in the Annex.
Guess first!

? What is the normal?

? From where are angles of incidence and reflection measured?

? State the two laws of reflection.

Reflection

Light bounces off a mirror by one simple rule.

PREDICT FIRST · NO CALCULATOR, JUST YOUR GUT

How tall does a mirror need to be to see all of yourself?

You are 160 cm tall and stand 2 m from a flat wall mirror. What is the shortest mirror that lets you see yourself from head to toe?

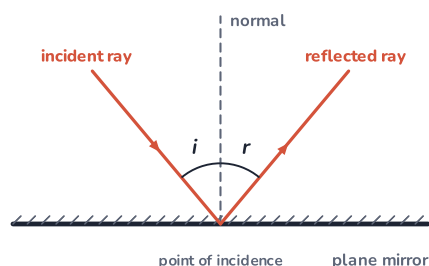
- ☐ A 160 cm, your full height
- ☐ B 80 cm, half your height
- ☐ C It depends on how far you stand from it

Why do you think so?

The **normal** is an imaginary line drawn at 90° to the surface at the point where the ray hits it. The **angle of incidence, i** is the angle between the incident ray and the normal. The **angle of reflection, r** is the angle between the reflected ray and the normal.

Laws of reflection

1. The angle of incidence equals the angle of reflection: $i = r$.
2. The incident ray, the reflected ray and the normal all lie in the **same plane**.



Both angles are measured from the **normal**, never from the mirror.

EXAMINER'S TIP

If a question gives the angle between the ray and the **mirror**, subtract it from 90° first.

A ray at 25° to the mirror has $i = 90^\circ - 25^\circ = 65^\circ$.

Worked example 1**[3]**

Try it in the box. Full solution in the Annex.

? List five features of an image in a plane mirror.

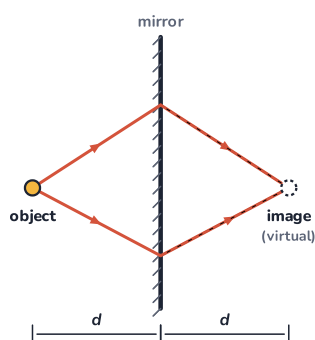
? What does virtual mean?

A ray of light strikes a plane mirror at an angle of 25° to the mirror surface. Find (a) the angle of incidence, (b) the angle of reflection, and (c) the angle between the incident ray and the reflected ray.

**The image in a plane mirror**

Rays from the object reflect off the mirror and spread out. Your eye traces them back in straight lines, so they seem to come from a point behind the mirror. No light actually goes there, so the image is **virtual**. It cannot be caught on a screen.

- **virtual** (cannot be formed on a screen)
- **upright** (the same way up as the object)
- **the same size** as the object
- **the same distance behind** the mirror as the object is in front
- **laterally inverted** (left and right are swapped)



The dashed lines are not real rays. They show where the reflected rays **appear** to come from.

DID YOU KNOW?**AMBULANCE**

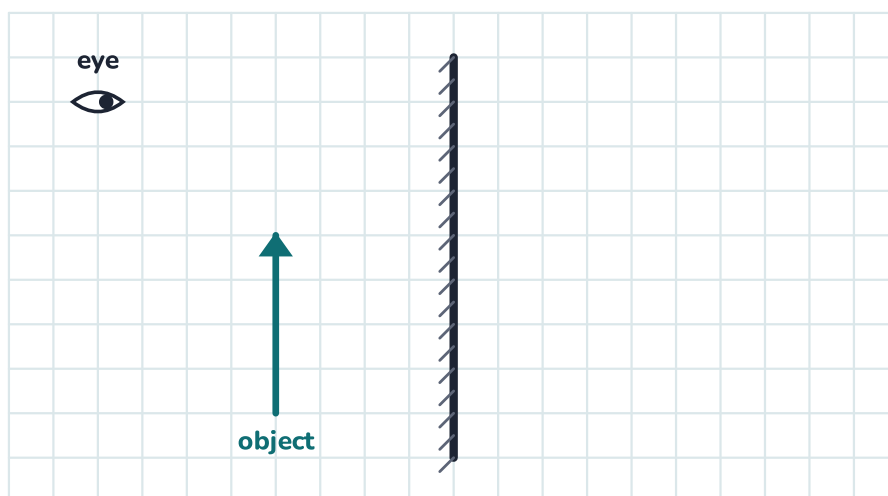
Ambulances have their name printed backwards on the front. In a driver's rear-view mirror, lateral inversion flips it back so it reads AMBULANCE.

Q1

Draw on the diagram.
Use a sharp pencil and
a ruler.

[4]

Construct the image of the object in the plane mirror. Then draw **two** rays that leave the top and the bottom of the object, reflect off the mirror and enter the eye.

**Q2**

Multiple choice. Tick
one box.

[1]

Which list describes the image formed by a plane mirror?

- ☐ **A** real, inverted, same size
- ☐ **B** virtual, upright, same size, laterally inverted
- ☐ **C** virtual, inverted, diminished
- ☐ **D** real, upright, magnified

Q3

Stretch

[3]

A ray of light hits a plane mirror. The ray stays fixed while the mirror is turned through 10° . Explain why the reflected ray turns through 20° .

02

Activity 1

? Define refraction.

? What causes refraction?

? When does a ray not bend at all?

Refraction

Light changes speed when it enters a new material, and bends.

TRY IT AT HOME

The coin that appears from nowhere

You need: an opaque mug, a coin, a jug of water, a friend

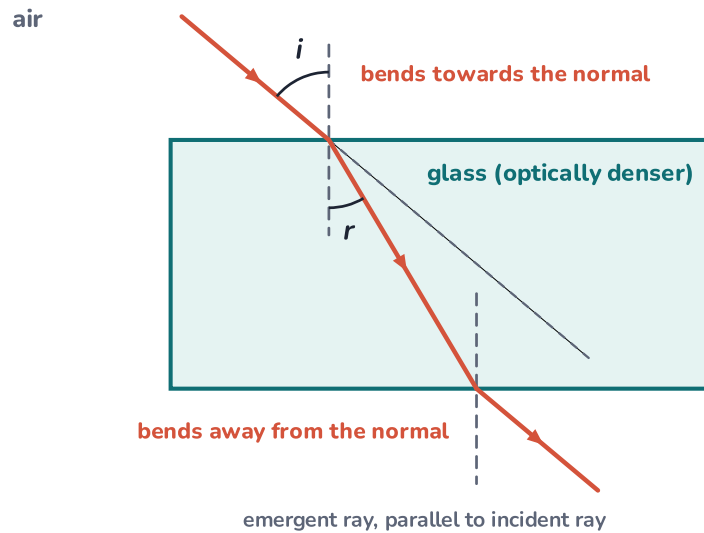
1. Put the coin at the bottom of the empty mug, near the side.
2. Move your head back until the rim of the mug just hides the coin. Keep your head still.
3. Ask your friend to pour water into the mug slowly, without moving the coin.

The coin comes back into view. Use the word refraction to explain why.

Refraction is the bending of light as it passes from one medium into another. It happens because light travels at **different speeds** in different materials.

A material in which light travels more slowly is called **optically denser**.

- Air → glass (light slows down): the ray bends **towards the normal**. The angle of refraction r is smaller than i .
- Glass → air (light speeds up): the ray bends **away from the normal**.
- A ray travelling **along the normal** ($i = 0^\circ$) does not bend. It still changes speed.



Drawn to scale for glass ($n = 1.50$) with $i = 50^\circ$, giving $r = 30.7^\circ$. The ray leaves **parallel** to the way it came in, shifted sideways.

? Why can n never be less than 1?

$$n = \sin i / \sin r$$

n refractive index of the medium no unit
 i angle in air (or a vacuum) $^\circ$
 r angle in the medium $^\circ$

This is Snell's law: for a given pair of materials, $\sin i / \sin r$ is a constant. n is always greater than 1.

? Define refractive index using speeds.

$$n = c / v$$

c speed of light in a vacuum $3.0 \times 10^8 \text{ m/s}$
 v speed of light in the medium m/s

Refractive index is the ratio of the speed of light in a vacuum to the speed of light in the medium. A bigger n means light is slowed down more.

REFRACTIVE INDEX AND THE SPEED OF LIGHT

Material	Refractive index n	Speed of light $v = c / n$
air	1.00 (almost exactly)	$3.0 \times 10^8 \text{ m/s}$
water	1.33	$2.3 \times 10^8 \text{ m/s}$
glass	1.50 (varies by type)	$2.0 \times 10^8 \text{ m/s}$
diamond	2.42	$1.2 \times 10^8 \text{ m/s}$

EXAMINER'S TIP · A CHECK THAT CATCHES MOST SLIPS

Whichever way the light is going, the **bigger angle is always in the air** and the smaller angle is in the glass or water. So $n = \sin(\text{angle in air}) \div \sin(\text{angle in the medium})$.

If you get $n < 1$, you have divided the wrong way round or used an angle measured from the surface.

Worked example 2**[4]**

Try it in the box. Full solution in the Annex.

A ray of light in air strikes a glass block at an angle of incidence of 50° . The refractive index of the glass is 1.50. Calculate (a) the angle of refraction, and (b) the speed of light in the glass.

Worked example 3**[3]**

Try it in the box. Full solution in the Annex.

A ray of light travelling **inside** a glass block ($n = 1.50$) meets the glass–air surface at an angle of incidence of 25° . Calculate the angle the ray makes with the normal after it leaves the glass.

Error hunt 1

Someone else's working. Find the slip.

A ray hits a glass block making an angle of 30° with the **surface** of the block. The angle of refraction is 35° . A student calculates the refractive index.

L1 $n = \sin i / \sin r$

L2 $n = \sin 30^\circ / \sin 35^\circ$

L3 $n = 0.500 / 0.574 = 0.87$

Circle the line with the mistake. Then write the correct working below.

Q4



Warm-up

[4]

The refractive index of water is 1.33.

(a) Calculate the speed of light in water.

[2]

(b) A ray in air strikes the water surface at an angle of incidence of 40° . Calculate the angle of refraction.

[2]

Q5

Multiple choice. Tick one box.

[1]

A ray of light enters a glass block along the normal. What happens to it?

- ☐ A It bends towards the normal and slows down.
- ☐ B It bends away from the normal and speeds up.
- ☐ C It does not bend, but it slows down.
- ☐ D It is totally internally reflected.

Q6



Stretch

[2]

Light passes from glass ($n = 1.50$) into water ($n = 1.33$). Does it bend towards or away from the normal? Explain your answer using the speed of light.

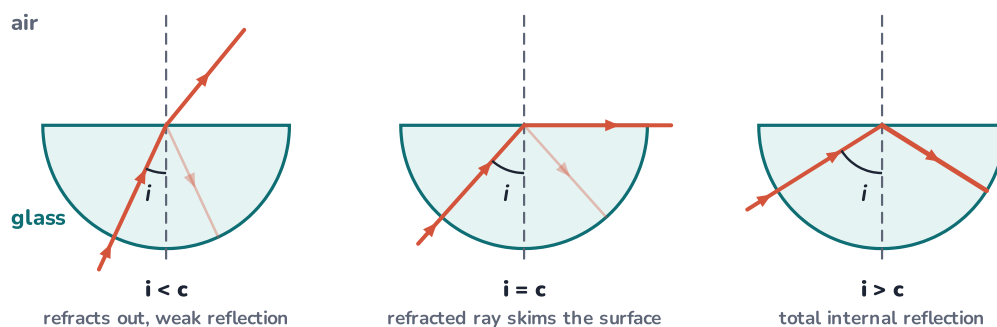
Total internal reflection

At a steep enough angle, light cannot escape. This is how optical fibres work.

Send light from glass into air and make the angle of incidence bigger. The refracted ray bends further and further from the normal. At one angle, the **critical angle c**, the refracted ray travels along the surface at 90° to the normal. Make the angle any bigger and no light escapes at all. It is **all** reflected back inside the glass. This is **total internal reflection (TIR)**.

03

- ? Define critical angle.
- ? State the two conditions for total internal reflection.



Glass with $n = 1.50$ has a critical angle of 41.8° . The three panels are drawn at 25° , 41.8° and 58° . A faint ray shows the weak partial reflection that happens alongside refraction.

? Can total internal reflection happen when light goes from air into water?

Total internal reflection needs BOTH

- light travelling from an **optically denser** medium towards a **less dense** one (glass $\rightarrow$ air, water $\rightarrow$ air)
- an angle of incidence **greater than the critical angle** ($i > c$)

It cannot happen when

- light goes from air into glass or water. It always refracts in
- i is less than c . Most light refracts out and a little is reflected

$$\sin c = 1 / n$$

c critical angle

n refractive index of the denser medium no unit

Where it comes from: at the critical angle the angle in air is 90° , so $n = \sin 90^\circ / \sin c = 1 / \sin c$.

Worked example 4

[5]

Try it in the box. Full solution in the Annex.

- (a) Calculate the critical angle for glass of refractive index 1.50. (b) Calculate the critical angle for diamond, $n = 2.42$. (c) Use your answers to suggest why a cut diamond sparkles much more than a glass copy.



Error hunt 2

Someone else's
working. Find the slip.

A student finds the critical angle for water, $n = 1.33$.

L1 $\sin c = 1 / n$

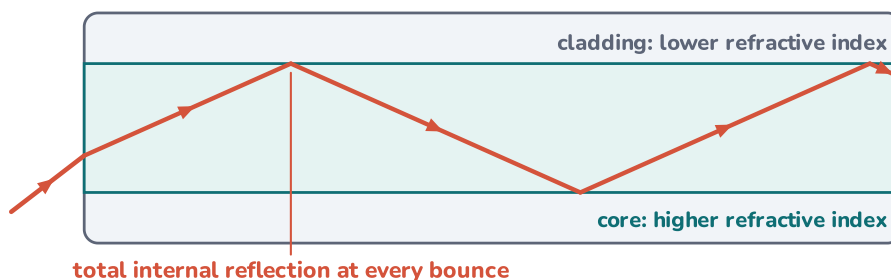
$$L2 \quad \sin c = 1,33 / 1 = 1,33$$

L3 $\sin^{-1}(1.33)$ gives Math ERROR, so water has no critical angle

Circle the line with the mistake. Then write the correct working below.

Optical fibres

An optical fibre is a very thin, flexible strand of glass. Its **core** is surrounded by **cladding** with a **lower refractive index**. Light entering one end hits the core–cladding boundary at an angle bigger than the critical angle, so it is totally internally reflected again and again. It follows the fibre even round gentle bends.



The light cannot leak out of the sides, so very little is lost even over many kilometres.

Telecommunication

- Internet, phone and TV signals travel as pulses of light.
- Carries **far more data** than a copper cable of the same size.
- **Less signal loss**, so fewer boosters are needed over long distances.
- **No electrical interference** and harder to tap into, so more secure.
- Thinner and lighter than copper for the same amount of data.

Medicine (the endoscope)

- A bundle of fibres carries light **into** the body to light it up.
- A second bundle carries the image **back out** to a camera.
- Doctors can see inside the stomach or lungs **without major surgery**.
- Small cuts mean less pain, less risk of infection and faster recovery.

Blank spaces are for your working. Solutions are in the Annex at the back.

? Why is the cladding given a lower refractive index than the core?

? Give three advantages of optical fibres over copper cables.

DID YOU KNOW?

Under the sea

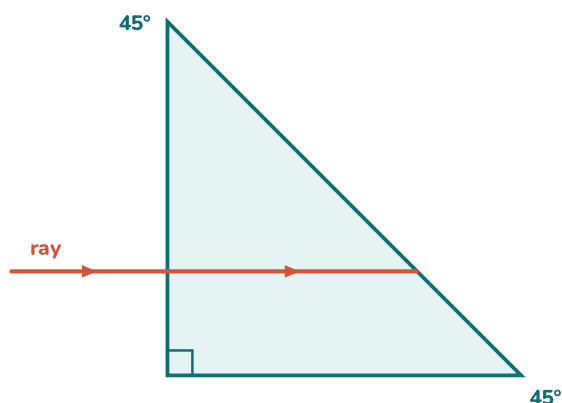
Almost all internet traffic between countries travels as light through optical fibre cables lying on the sea floor. Singapore is one of the busiest landing points for these cables in the world.

Q7

Draw on the diagram.
Use a sharp pencil and
a ruler.

[5]

A ray of light enters a right-angled glass prism at 90° to face AB, as shown. The refractive index of the glass is 1.50.



(a) Explain why the ray does not bend as it enters the prism. (b) Show that the ray is totally internally reflected at the long face. (c) Complete the path of the ray until it leaves the prism.

Q8

Multiple choice. Tick
one box.

[1]

Which is **not** an advantage of optical fibres over copper cables for carrying phone signals?

- ☐ A They carry much more data.
- ☐ B They lose less signal over long distances.
- ☐ C They are not affected by electrical interference.
- ☐ D They can carry electrical power to the phone at the same time.

04

? Define the principal focus.

? Define focal length.

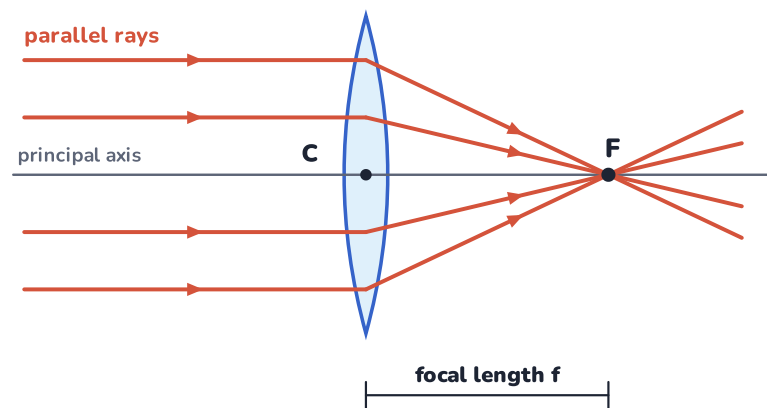
? Name the three rays you can draw without measuring any angles.

Thin converging lenses

How a curved piece of glass forms images.

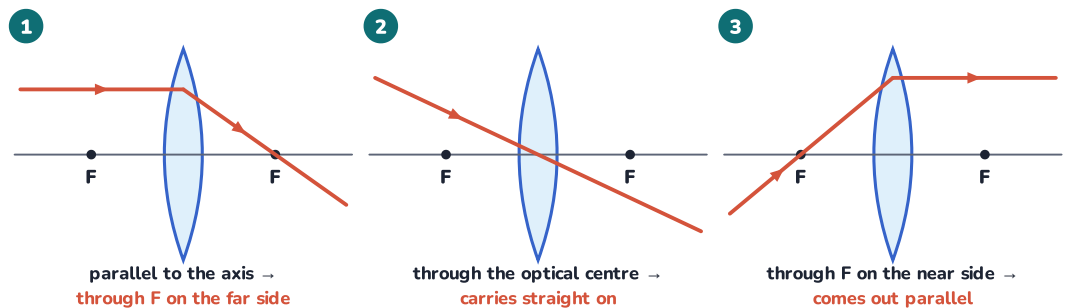
A **converging lens** (convex lens) is thicker in the middle than at the edges. It refracts rays so that they come together.

- **Principal axis:** the line through the centre of the lens, at 90° to it.
- **Optical centre, C:** the centre of the lens. Rays through it pass straight through.
- **Principal focus, F:** the point where rays parallel to the principal axis converge after passing through the lens. There is one on each side.
- **Focal length, f:** the distance between the optical centre and the principal focus.



A thicker, more curved lens bends light more, so it has a shorter focal length.

The three construction rays



Any two of these rays are enough to find the image. Ray 2 is the easiest to draw accurately.

? How do you find a virtual image on a ray diagram?

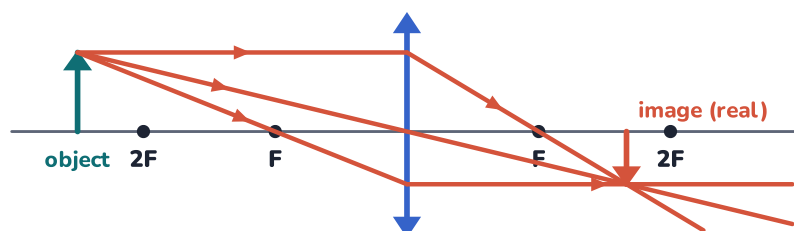
How to draw a ray diagram

1. Choose a scale. Draw the principal axis and the lens as a vertical line with arrowheads.
2. Mark F and 2F on **both** sides of the lens, using the scale.
3. Draw the object as an upright arrow standing on the axis.
4. From the tip of the object draw ray 1 (parallel, then through F) and ray 2 (straight through C).
5. Where the rays **cross** is the tip of the image. Draw the image arrow from the axis to that point.
6. If the rays **spread apart** after the lens, extend them backwards with dashed lines. Where the dashed lines meet is a **virtual** image.

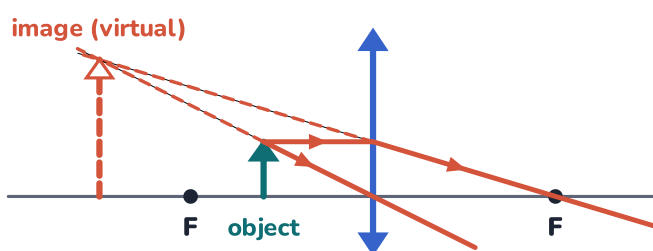
EXAMINER'S TIP

Marks go for: a **ruler**, **arrows** on every ray, real rays **solid**, virtual extensions **dashed**, and the image clearly **labelled**.

Describe images with three words from three pairs: **real / virtual**, **upright / inverted**, **magnified / diminished / same size**.



Object beyond 2F: the rays really meet, so the image is **real**. It is inverted and diminished. This is how a camera and your eye work.



Object between F and the lens: the rays spread out and only **seem** to meet, so the image is **virtual**, upright and magnified.

? Which object position gives a same-size image?

? Which position does a magnifying glass use?

WHERE THE IMAGE FORMS · LEARN THIS TABLE

Object position	Image position	Nature of image	Used in
very far away	at F on the other side	real, inverted, diminished	focusing on distant objects
beyond 2F	between F and 2F on the other side	real, inverted, diminished	camera, the eye
at 2F	at 2F on the other side	real, inverted, same size	photocopier (same-size copy)
between F and 2F	beyond 2F on the other side	real, inverted, magnified	projector
at F	no image: rays come out parallel	—	spotlight, torch beam
between F and the lens	same side as the object, further away	virtual, upright, magnified	magnifying glass

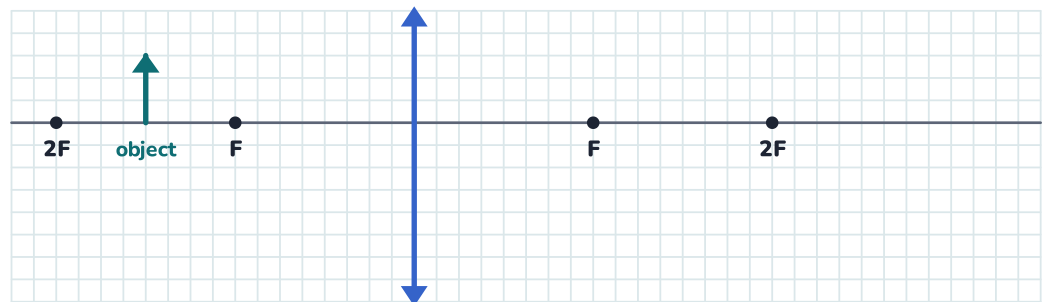
Real images form on the opposite side of the lens and can be caught on a screen. The virtual image is on the same side as the object.

Worked example 5

[5]

Try it in the box. Full solution in the Annex.

An object 3.0 cm tall stands 12 cm from a thin converging lens of focal length 8.0 cm. By drawing a ray diagram to scale on the grid, find the position and height of the image, and describe it.



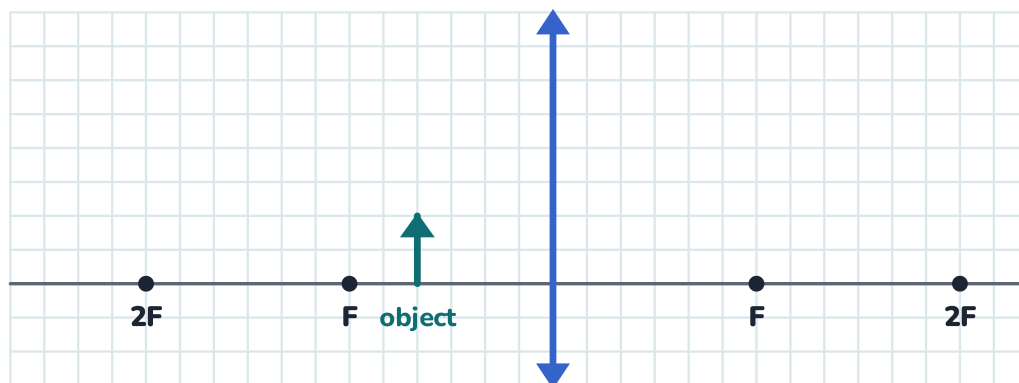
Scale: 1 square = 1 cm

Q9

Draw on the diagram.
Use a sharp pencil and
a ruler.

[5]

A magnifying glass has a focal length of 6.0 cm. It is held 4.0 cm from a beetle 2.0 cm tall. Complete the ray diagram to scale to find the image.



Scale: 1 square = 1 cm

State the distance of the image from the lens, its height, and three features of the image.

Q10

Multiple choice. Tick
one box.

[1]

A student uses a converging lens as a magnifying glass. Where must the object be placed?

- | | |
|--|--|
| ☐ A beyond 2F | ☐ B at 2F |
| ☐ C between F and 2F | ☐ D between F and the lens |

Q11

Multiple choice. Tick
one box.

[1]

An object is placed at 2F in front of a converging lens. Which describes the image?

- ☐ **A** virtual, upright, same size
- ☐ **B** real, inverted, same size
- ☐ **C** real, inverted, magnified
- ☐ **D** real, upright, diminished

Q12

Exam

[5]

A camera uses a converging lens to form an image of a distant tree on the light sensor.

(a) State three features of the image formed on the sensor. **[2]**

(b) Explain why the image must be real, not virtual. **[1]**

(c) The camera is moved closer to the tree. Does the sensor need to move further from, or closer to, the lens? Explain using the table of image positions. **[2]**

DID YOU KNOW?

Upside down

The lens in your eye forms a real, inverted, diminished image on your retina, just like a camera. Your brain turns it the right way up. Newborn babies are still learning to make sense of it.

05

Pull it together

Check yourself before you move on.

KEY WORDS

normal

line at 90° to a surface where a ray hits it

refraction

bending of light as it passes from one medium into another

refractive index

$$n = c / v = \sin i / \sin r$$

critical angle

angle of incidence in the denser medium that gives a refracted angle of 90°

total internal reflection

all light reflected back when $i > c$, going from denser to less dense

principal focus

point where rays parallel to the principal axis converge

focal length

distance from the optical centre to the principal focus

real image

formed where rays actually meet; can be caught on a screen

virtual image

where rays only appear to come from; cannot be caught on a screen

Q13

Multiple choice. Tick one box.

[1]

Light travels from glass into air. The angle of incidence is increased steadily from 0° . Which describes what happens at the moment the angle of incidence becomes greater than the critical angle?

- ☐ **A** The refracted ray bends towards the normal.
- ☐ **B** The refracted ray disappears and all the light is reflected inside the glass.
- ☐ **C** The light passes straight through without bending.
- ☐ **D** The reflected ray disappears.

SUMMARY · EXPLAIN TO A FRIEND WHY THE STRAW IN YOUR DRINK LOOKS BENT AT THE WATER SURFACE

HOW CONFIDENT ARE YOU? TICK ONE, THEN DIARY THE REVIEW DATE

☐ **Shaky**

Redo the cue questions tomorrow.

Review on ____ / ____

☐ **Getting there**

Redo the worked examples in 3 days.

Review on ____ / ____

☐ **Nailed it**

Cue questions only, in a week.

Review on ____ / ____

A

Annex · Solutions

Try every question on your own first. For drawings, lay your answer next to the one here and check the image position within ± 1 small square.

Predict 1

1 · Reflection

80 cm, half your height, at any distance.

Light from your feet reaches your eye by bouncing off the mirror at a point halfway between your eye level and your feet ($i = r$). Light from the top of your head bounces halfway between your eye and the top of your head.

So the mirror only has to cover half of you. Step back and both halfway points stay where they are, so distance makes no difference.

Worked example 1

- 1 (a) Angles are measured from the normal, which is at 90° to the mirror.

$$i = 90^\circ - 25^\circ = 65^\circ$$

- 2 (b) Law of reflection.

$$r = i = 65^\circ$$

- 3 (c) The two rays sit either side of the normal.

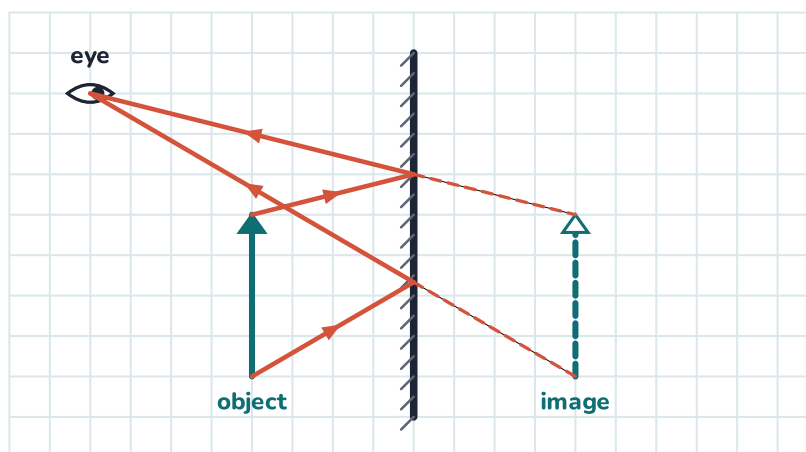
$$\text{angle between rays} = i + r = 65^\circ + 65^\circ = 130^\circ$$

Sketch it. 50° is the common wrong answer, from using $25^\circ + 25^\circ$.

Answer: **(a) 65° (b) 65° (c) 130°**

Q1 · drawing

- 1 Mark the image the same distance behind the mirror as the object, the same size and upright.
object is 4 squares in front, so image is 4 squares behind
- 2 Join the image to the eye with a straight line.
Where it crosses the mirror is where the real ray reflects.
draw it dashed behind the mirror
- 3 Draw the real ray: object → that point on the mirror → eye. Add arrows.
repeat for the bottom of the object

**Q2 · MCQ****B**

A plane mirror image cannot be caught on a screen (virtual), is the right way up, the same size, and left-right reversed.

Q3

The normal is fixed to the mirror, so it also turns through **10°**. The angle of incidence therefore changes by 10°.

Since $r = i$, the angle of reflection also changes by 10°, on the other side of the normal, which has itself moved 10°.

The two changes add, so the reflected ray turns through **10° + 10° = 20°**.

2 · Refraction**Activity 1**

Light from the coin **refracts away from the normal** as it leaves the water and enters the air, because it speeds up.

The bent ray now clears the rim and reaches your eye. Your brain assumes light travels in straight lines, so it traces the ray back and sees the coin **higher up** than it really is.

Worked example 2

1 (a) Rearrange $n = \sin i / \sin r$.
 $\sin r = \sin i / n = \sin 50^\circ / 1.50 = 0.766 / 1.50 = 0.511$

2 Take the inverse sine. Check your calculator is in degrees.
 $r = \sin^{-1}(0.511) = 30.7^\circ$

r is smaller than i . The ray bent towards the normal, as it should.

3 (b) Use $n = c / v$.
 $v = c / n = 3.0 \times 10^8 / 1.50 = 2.0 \times 10^8 \text{ m/s}$

Answer: **(a) 30.7° (b) $2.0 \times 10^8 \text{ m/s}$**

Worked example 3

1 The larger angle is in the air, so the 25° goes on the bottom.
 $n = \sin \theta_{\text{air}} / \sin \theta_{\text{glass}}$, so $1.50 = \sin \theta_{\text{air}} / \sin 25^\circ$

Label the angles by material, not by i and r .

2 Rearrange.
 $\sin \theta_{\text{air}} = 1.50 \times \sin 25^\circ = 1.50 \times 0.423 = 0.634$

3 Inverse sine.
 $\theta_{\text{air}} = 39.3^\circ$

Answer: **39.3° (bent away from the normal)**

Error hunt 1

L2 is wrong. 30° is measured from the surface, not the normal. The angle of incidence is $i = 90^\circ - 30^\circ = 60^\circ$.

$$n = \sin 60^\circ / \sin 35^\circ = 0.866 / 0.574 = \mathbf{1.51}.$$

Sanity check: $n = 0.87$ is impossible. Nothing slows light down to *faster* than in a vacuum.

Q4

(a)

$$v = c / n = 3.0 \times 10^8 / 1.33 = \mathbf{2.26 \times 10^8 \text{ m/s}}$$

(b)

1 Rearrange Snell's law.
 $\sin r = \sin 40^\circ / 1.33 = 0.643 / 1.33 = \mathbf{0.483}$

2 Inverse sine.
 $r = \mathbf{28.9^\circ}$

Q5 · MCQ

C

Along the normal, $i = 0^\circ$, so $r = 0^\circ$: no bending. The speed still drops, because the light is now in glass.

Q6

Away from the normal.

Water has a lower refractive index than glass, so light travels **faster** in water (2.26×10^8 m/s) than in glass (2.0×10^8 m/s). Light that speeds up bends away from the normal.

3 · Total internal reflection**Worked example 4**

1 (a) Use $\sin c = 1 / n$.
 $\sin c = 1 / 1.50 = 0.667$, so $c = 41.8^\circ$

2 (b) Same for diamond.
 $\sin c = 1 / 2.42 = 0.413$, so $c = 24.4^\circ$

3 (c) Compare.
diamond's critical angle is much smaller

Name the idea: total internal reflection.

Light inside the diamond only needs to hit a face at more than 24.4° to be totally internally reflected, so it bounces around inside many times before coming out through the top. The cut is designed to send it back out towards your eye.

Answer: **(a) 41.8° (b) 24.4°**

Error hunt 2

L2 is wrong. The fraction was turned upside down: $\sin c = 1 / 1.33$, not $1.33 / 1$.
 $\sin c = 1 / 1.33 = 0.752$, so $c = 48.8^\circ$.

Sanity check: a sine can never be bigger than 1. A Math ERROR is a clue that you have divided the wrong way.

Q7 · drawing

(a) The ray travels **along the normal** to the face ($i = 0^\circ$), so it does not change direction.

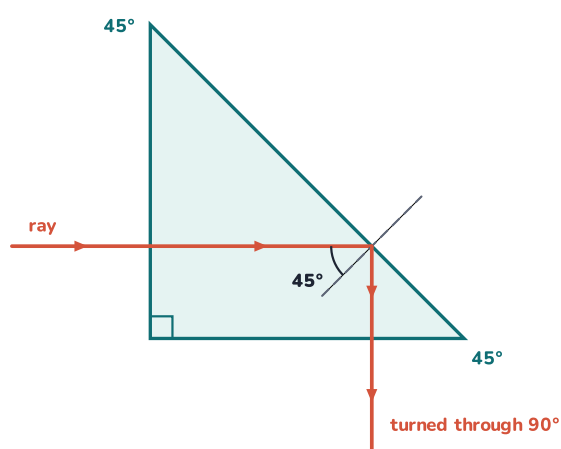
- 1 (b) Find the critical angle.
 $\sin c = 1 / 1.50$, so $c = 41.8^\circ$

- 2 The ray meets the long face at an angle of incidence of 45° .

State both numbers and compare them.

$45^\circ > 41.8^\circ$, so **total internal reflection happens**

(c) The ray is turned through 90° and leaves through the bottom face along the normal, without bending.



Prisms used like this replace mirrors in periscopes and binoculars. They reflect all the light and never tarnish.

Q8 · MCQ

D

Glass fibres carry light, not electric current. A, B and C are the real advantages.

**Worked example
5****4 · Thin converging lenses**

- 1 Mark F at 8 squares and 2F at 16 squares on both sides. The object is drawn 12 squares left of the lens.

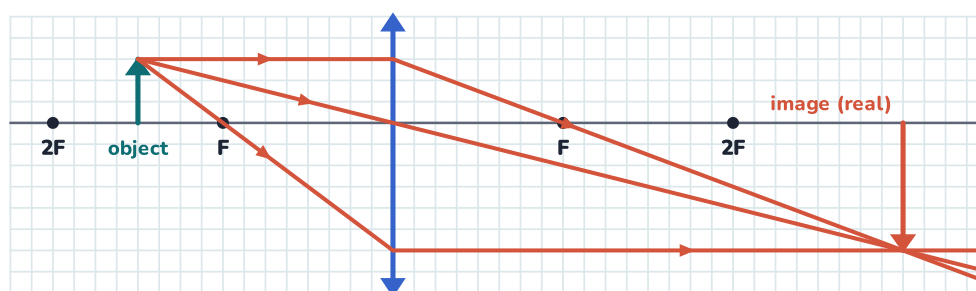
object is between F and 2F

- 2 Draw ray 1 from the object tip parallel to the axis, then through F on the far side. Draw ray 2 straight through C.

the rays cross 24 cm right of the lens, 6.0 cm below the axis

- 3 A third ray through the near F comes out parallel and meets the other two at the same point. It is a good check.

image tip at (24 cm, -6.0 cm)



Scale: 1 square = 1 cm

The image is **24 cm** from the lens on the opposite side and **6.0 cm** tall. It is **real, inverted and magnified** (twice the object's height). A careful drawing within ± 1 cm earns full marks.

Answer: **24 cm from the lens, 6.0 cm tall; real, inverted, magnified**

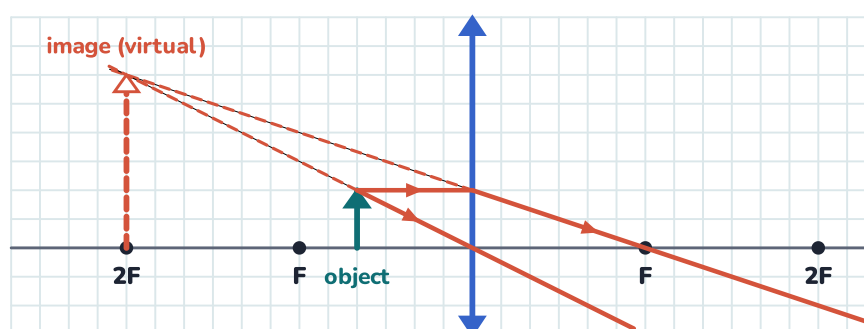
Q9 · drawing

- 1 Ray 1 from the tip goes parallel to the axis, then through F on the far side. Ray 2 goes straight through C.

the two rays spread apart after the lens

- 2 Extend both rays backwards with dashed lines until they meet.

they meet 12 cm left of the lens, 6.0 cm above the axis



Scale: 1 square = 1 cm

The image is **12 cm** from the lens, on the **same side** as the beetle, and **6.0 cm** tall. It is **virtual, upright and magnified** (3 times).

Q10 · MCQ

D

Only an object closer to the lens than F gives an upright, magnified, virtual image.

Q11 · MCQ

B

At 2F the image forms at 2F on the other side: real, inverted and the same size.

Q12

(a)

Real, inverted, diminished. (All three for 2 marks, two for 1 mark.)

(b)

Only a **real** image, where rays actually meet, can be **formed on a screen or sensor**. A virtual image has no light arriving at it.

(c)

Further from the lens. As the object moves closer (towards 2F), the image moves further from the lens (from F towards 2F), so the sensor must move back to keep the image sharp.

Q13 · MCQ

5 · Pull it together**B**

Beyond the critical angle no light can refract out. All of it is reflected: total internal reflection.